

VNU-HCMUS

Course name: Calculus III	Course code: MTH253
Time: 90 minutes	Date: 21/06/2024
Note: Students are [allowed / not allowed] to use materials during the examination.	

Full name of Student: _____ **Student ID:** _____ **No:** _____

Question 1 (2 points).

Find the limit, if it exists, or show that the limit does not exist:

- a) $\lim_{(x,y) \rightarrow (-1,2)} \frac{x^2 y^3}{x^2 + 2y^2}$
- b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^3}{x^2 + 2y^2}$
- c) For which value of a the following function is continuous at $(0,0)$:

$$f(x, y) = \begin{cases} \frac{x^2 y^3}{x^2 + 2y^2} & \text{if } (x, y) \neq (0, 0) \\ a & \text{if } (x, y) = (0, 0) \end{cases}$$

Question 2 (3 points).

- a) If $f(x, y) = x^2 \sin(y) + ye^{xy}$, where $x = s + 2t$ and $y = st$, use the Chain Rule to find $\frac{\partial f}{\partial s}$ and $\frac{\partial f}{\partial t}$ when $s = 0$ and $t = 1$.
- b) If $x^2 - 3yz^2 + xyz - 2 = 0$ then find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
- c) Find the directional derivative of the function $f(x, y) = e^x \sin(y)$ at the point $(0, \frac{\pi}{3})$ in the direction of the vector $v = \langle -6, 8 \rangle$.

Question 3 (3 points).

Find the extreme values of the function $f(x, y) = x^2 + y^2 + 4x - 4y$ on the disk $x^2 + y^2 \leq 9$.

Question 4 (2 points).

Find the volume of the solid under the surface $z = 1 + x^2 y^2$ and above the region enclosed by $x = y^2$ and $x = 4$.